

ExamEasy KENYA - STUDENT RESOURCE PORTAL

CONFIDENTIAL & VERIFIED ACADEMIC RESOURCE

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FACULTY: SCHOOL OF COMPUTING & INFORMATICS

COURSE UNIT: BIT 301 - Database Management Systems

EXAMINATION TYPE: PAST PAPER

ACADEMIC YEAR: 2024 | SEMESTER 1

PRICE: KES 200.00 (PAID & VERIFIED)

SOLUTIONS: COMPREHENSIVE MARKING SCHEME INCLUDED

TOTAL PAGES: 5 PAGES | WATERMARK: EXAMEASY GENUINE COPY

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INSTRUCTIONS TO CANDIDATES:

1. Answer Question ONE in Section A (Compulsory) and any other TWO questions in Section B.
2. Write your answers legibly in the provided answer booklet.
3. Non-programmable scientific calculators are permitted.
4. Time Allowed: 2 Hours 30 Minutes.

SECTION A: COMPULSORY (30 MARKS)

1. Given relation $R(A, B, C, D, E)$ with FDs $\{A \rightarrow B, BC \rightarrow D, D \rightarrow E\}$, determine candidate keys and highest normal form [10 Marks]
2. Write advanced SQL queries with window functions (RANK, PARTITION BY) for an e-commerce database [10 Marks]

SECTION B: ANSWER ANY TWO QUESTIONS (20 MARKS EACH)

2. (a) Formulate the optimization function and solve for boundary constraints. [10 Marks]
2. (b) Provide step-by-step analytical derivation with full annotations. [10 Marks]
3. (a) Evaluate real-world case studies demonstrating strategic failure and remediation. [12 Marks]
3. (b) Propose architectural best practices for scalable deployment. [8 Marks]

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DETAILED MARKING SCHEME & MODEL SOLUTIONS:

- Question 1 Model Solution with scoring rubrics and step breakdown.
 - Complete worked steps verified by university external examiners.
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